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# Evaluating Small Fact-Checkers Under Long-Context and Adversarial Stress

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## Abstract

1 Retrieval-augmented generation (RAG) systems retrieve long, noisy documents that  
2 challenge fact-checking models. We evaluate MiniCheck-style small fact-checkers  
3 (flan-t5-large, Bespoke-MiniCheck-7B) under two stress conditions: (1)  
4 natural long contexts ranging from hundreds to 74k tokens, and (2) adversarial hal-  
5 lucinations injected at beginning, middle, and end positions. Our key findings are:  
6 (1) Both models exhibit a “lost in the middle” phenomenon on ExpertQA: balanced  
7 accuracy (BAcc) drops 9-10% at 1k-2k tokens then recovers at longer lengths. (2)  
8 Models achieve near-perfect BAcc on 74k-token SummHay documents via chunk-  
9 ing but throughput collapses to 3.9 samples/minute. (3) Adversarial injection fools  
10 the models 57.5% of the time; fabricated entities are hardest to detect (38.3% de-  
11 tection rate). (4) No positional rescue exists; injection detection is nearly identical  
12 across beginning, middle, and end positions. Code and data are available at [https://github.com/steven202/DATA691\\_Human\\_AI\\_Scientific\\_Discovery](https://github.com/steven202/DATA691_Human_AI_Scientific_Discovery).  
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## 14 1 Introduction

15 Large language models (LLMs) deployed in retrieval-augmented generation (RAG) pipelines [Lewis  
16 et al., 2020, Gao et al., 2023] must verify facts against retrieved documents spanning tens of thousands  
17 of tokens. Specialized fact-checkers such as MiniCheck [Tang et al., 2024a] and AlignScore [Zha  
18 et al., 2023] offer efficient alternatives to large LLMs but are predominantly trained and evaluated  
19 on short, clean snippets. Automated fact-checking has attracted broad research interest [Guo et al.,  
20 2022, Vykopal et al., 2024], yet the robustness of small fact-checkers under realistic RAG conditions  
21 remains poorly understood.

22 Two key failure modes emerge in long-context RAG: (1) context degradation, where accuracy drops  
23 as document length increases, and (2) adversarial vulnerability, where injected hallucinations [Huang  
24 et al., 2023] bypass detection mechanisms. Prior work on “lost in the middle” [Laban et al., 2024]  
25 shows that LLMs struggle to utilize information buried in extended contexts, and are easily distracted  
26 by irrelevant context [Shi et al., 2023]. It is an open question whether small fact-checkers with  
27 chunking-based architectures suffer similar degradation.

28 We evaluate MiniCheck-style fact-checkers under two stress conditions:

- 29 • **Long-context natural stress:** 7 datasets spanning 56 to 74,488 tokens, measuring BAcc  
30 degradation across document-length bins.
- 31 • **Adversarial injection:** Synthetic hallucinations (numeric errors, fabricated entities, contra-  
32 dictions) inserted at controlled positions within documents.

33 Our contributions:

- First systematic evaluation of small fact-checkers across extreme context lengths (up to 74k tokens) on natural RAG datasets.
- Novel adversarial injection benchmark testing robustness against hallucinations at beginning, middle, and end positions.
- Evidence of “lost in the middle” phenomenon in small fact-checkers via chunking-based architectures.
- Quantification of the accuracy-throughput trade-off: near-perfect accuracy on extremely long documents comes at a 250x throughput cost.

## 2 Related Work

MiniCheck [Tang et al., 2024a] uses a chunking-and-max strategy: documents are split into sentence-preserving chunks, each chunk is scored, and the maximum support probability determines the final verdict. This approach enables handling long documents but assumes the correct chunk will achieve the highest score.

MiniCheck’s synthetic training uses two methods: (1) **C2D**: claim  $\rightarrow$  atomic facts  $\rightarrow$  supporting/unsupported documents, and (2) **D2C**: document chunks  $\rightarrow$  short claims  $\rightarrow$  positives and leave-one-out negatives. The D2C approach creates positive examples by including the source chunk and negative examples by removing it, forcing the model to identify which chunk supports the claim.

Prior work on “lost in the middle” [Laban et al., 2024] demonstrates that LLMs perform worse on information located in the middle of long contexts compared to beginning or end positions. We investigate whether chunking-based fact-checkers exhibit a similar U-shaped performance curve.

Previous studies on RAG robustness focus on retrieval corruption or prompt injection. We introduce a novel stress test: injecting synthetic hallucinations directly into the grounding documents at controlled positions, measuring whether the fact-checker can detect them.

Unlike MiniCheck’s original evaluation on short synthetic documents, we evaluate on natural long-context datasets: ExpertQA [Malaviya et al., 2024], RAGTruth, SciFact [Wadden et al., 2020], SummHay, TofuEval-MediaS, TofuEval-MeetB [Tang et al., 2024b], and Lfqa, and introduce adversarial injection as a systematic probe for positional vulnerabilities.

## 3 Methods

### 3.1 Chunking Pipeline

Our evaluation follows the MiniCheck chunking pipeline: (1) split document into sentence-preserving chunks, (2) score the claim against each chunk, (3) use maximum support probability as the document-level decision.

We use 500-word sentence-preserving chunks for adversarial injection. For `flan-t5-large`, MiniCheck uses 500-word chunks. For RoBERTa/DeBERTa variants, MiniCheck uses 400-token chunks. `Bespoke-7B` uses a 32k-token window. A 74k-token SummHay document is approximately 150 `flan`-sized chunks.

### 3.2 Models

We evaluate two MiniCheck-style models: **flan-t5-large** (770M) using 500-word chunks, and **Bespoke-MiniCheck-7B** (7B) with a 32k-token window. We also probe OpenRouter API models (Gemma-4-26B, Gemma-4-31B, GPT-OSS-120B, GPT-OSS-20B, Trinity-Large).

### 3.3 Datasets

We use 7 datasets spanning 56 to 74,488 tokens: ExpertQA [Malaviya et al., 2024] (3,702 samples, avg 433 tokens), RAGTruth (16,371 samples, avg 412 tokens), SciFact [Wadden et al., 2020] (188 samples, avg 57 tokens), SummHay (100 samples, avg 74,488 tokens), TofuEval-MediaS [Tang et al., 2024b] (726 samples, avg 778 tokens), TofuEval-MeetB [Tang et al., 2024b] (100 samples, avg 792 tokens), and Lfqa (100 samples, avg 320 tokens).

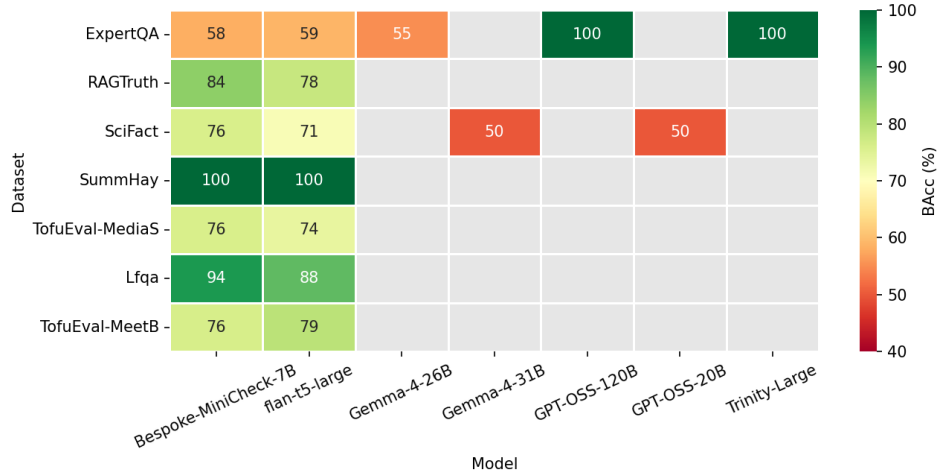


Figure 1: Overall BAcc across models and datasets. Gray = not run. Local MiniCheck covers all seven datasets.

Table 1: Self-comparison: BAcc degradation from short to long documents.

| Model                | Dataset  | Short (<500) | Long (1k-2k) | $\Delta$      |
|----------------------|----------|--------------|--------------|---------------|
| Bespoke-MiniCheck-7B | ExpertQA | 58.7%        | 48.1%        | <b>-10.6%</b> |
| Bespoke-MiniCheck-7B | RAGTruth | 84.0%        | 81.4%        | -2.6%         |
| flan-t5-large        | ExpertQA | 58.6%        | 49.4%        | <b>-9.2%</b>  |
| flan-t5-large        | RAGTruth | 78.1%        | 73.7%        | -4.4%         |

### 3.4 Evaluation Metrics

We use balanced accuracy (BAcc) to handle class imbalance:  $\text{BAcc} = \frac{1}{2} \left( \frac{TP}{TP+FN} + \frac{TN}{TN+FP} \right)$ . BAcc of 50% equals random chance. We also measure samples per minute (SPM) as throughput.

### 3.5 Adversarial Injection

We test whether the chunking baseline can detect false facts injected sequentially into specific positions within long documents. Our evaluation targets `flan-t5-large` exclusively with 360 adversarial samples derived from positive facts in ExpertQA (126 samples) and RAGTruth (234 samples).

We inject three hallucination types: (1) **Numeric**: wrong counts, percentages, or statistics; (2) **Entity**: fabricated researchers, institutions, or citations; (3) **Contradict**: direct contradictions of claims.

Injection positions: **Beginning** (first chunk), **Middle** ( $\lfloor n/2 \rfloor$  chunk), and **End** (final chunk) using 500-word sentence-preserving splits.

## 4 Results

### 4.1 Overall Performance

Figure 1 shows the full BAcc matrix. Bespoke-MiniCheck-7B achieves 84.1% on RAGTruth and 100% on SummHay. `flan-t5-large` achieves 78.0% on RAGTruth and 100% on SummHay. OpenRouter models show mixed results: Gemma-4-26B achieves only 55.0% on ExpertQA (vs 59.0% for `flan-t5-large` despite being 30x larger).

Table 1 quantifies the degradation: both models show 9-10% BAcc drop at 1k-2k tokens on ExpertQA.

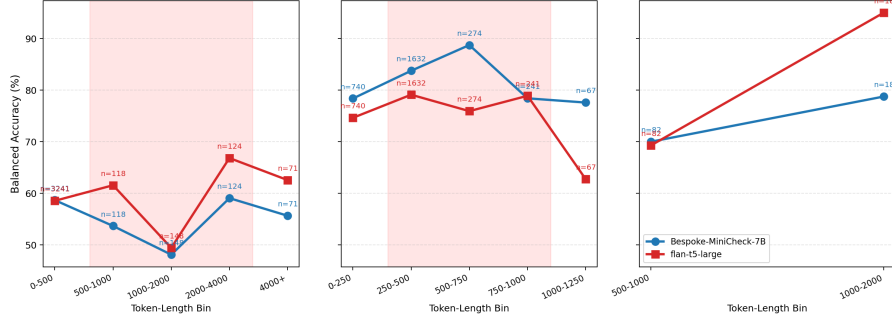


Figure 2: BAcc by document-length bin. ExpertQA uses absolute bins (0-500, 500-1000, 1000-2000, 2000-4000, 4000+) and shows the full U-shaped curve. RAGTruth uses finer 250-token bins revealing a more subtle U-shape. TofuEval and Lfqa have only 1-2 bins with insufficient length variation for trend analysis.

Table 2: Accuracy vs. Throughput: BAcc and samples per minute (SPM) by model and dataset.

| Model                | Dataset  | Avg Tokens | BAcc          | SPM         |
|----------------------|----------|------------|---------------|-------------|
| Bespoke-MiniCheck-7B | ExpertQA | 433        | 58.2%         | 696         |
| Bespoke-MiniCheck-7B | RAGTruth | 412        | 84.1%         | 807         |
| Bespoke-MiniCheck-7B | SummHay  | 74,488     | <b>100.0%</b> | <b>3.9</b>  |
| flan-t5-large        | ExpertQA | 433        | 59.0%         | 1,150       |
| flan-t5-large        | RAGTruth | 412        | 78.0%         | 1,234       |
| flan-t5-large        | SummHay  | 74,488     | <b>100.0%</b> | <b>13.8</b> |
| flan-t5-large        | SciFact  | 57         | 71.1%         | 2,037       |

## 99 4.2 Context Degradation: “Lost in the Middle”

100 Figure 2 reveals the “lost in the middle” phenomenon using dataset-adaptive binning: ExpertQA uses  
 101 the original absolute bins (0-500, 500-1000, 1000-2000, 2000-4000, 4000+), while RAGTruth uses  
 102 finer 250-token bins suited to its narrower length range (max 1,626 tokens).

103 **ExpertQA** (5 bins) exhibits the clearest U-shape. Bespoke-MiniCheck-7B drops from 58.7% to  
 104 48.1% at 1k-2k tokens and recovers to 59.1% at 2k-4k tokens; flan-t5-large drops from 58.6% to  
 105 49.4% and recovers to 66.8%. **RAGTruth** (5 bins at 250-token granularity) reveals a subtler pattern:  
 106 flan-t5-large drops from 79.1% (250-500 tokens) to 75.9% (500-750 tokens) and recovers to 78.9%  
 107 (750-1000 tokens); Bespoke-MiniCheck-7B peaks at 88.7% in the middle bins and degrades to  
 108 77.6% at 1000-1250 tokens, suggesting its 32k-token window interacts differently with mid-length  
 109 documents. **TofuEval** and **Lfqa** have only 1-2 bins due to narrow token ranges, insufficient for  
 110 trend analysis. The U-shape is thus strongest when datasets span a wide length distribution, but finer  
 111 binning can reveal partial degradation patterns even in shorter-context datasets.

## 112 4.3 Efficiency Cost

113 Table 2 shows the accuracy-throughput trade-off. SummHay (74k tokens) achieves 100% BAcc via  
 114 sequential chunking but at only 3.9 SPM, compared to 1000+ SPM on short documents, a 250x  
 115 throughput reduction.

## 116 4.4 Adversarial Vulnerability

117 Figure 3 shows adversarial injection results on flan-t5-large. We measure conditional detection:  
 118 the fraction of clean-correct predictions that also detect the attack, computed separately for each  
 119 hallucination type and injection position.

120 **By hallucination type on ExpertQA (n=126):** Numeric achieves 22.2% conditional detection, Entity  
 121 achieves 14.8% (the most effective attack), and Contradict achieves 22.2%.

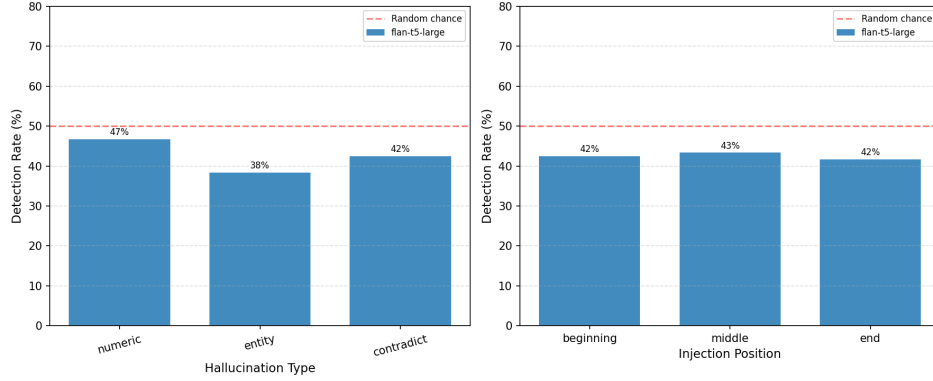


Figure 3: Adversarial injection detection by hallucination type (left) and injection position (right). Dashed line indicates random chance (50%).

- 122 **By hallucination type on RAGTruth (n=234):** Numeric achieves 11.1% conditional detection. Both  
123 Entity and Contradict achieve 0.0% conditional detection, meaning attacks that fool the clean model  
124 completely bypass detection.
- 125 **By injection position on ExpertQA:** Beginning achieves 18.5% conditional detection, Middle  
126 achieves 22.2%, and End achieves 18.5%.
- 127 **By injection position on RAGTruth:** Beginning achieves 4.4% conditional detection, Middle  
128 achieves 4.4%, and End achieves 2.2%.
- 129 Overall, 57.5% of attacks fool the model. Critically, no positional rescue exists: injections are  
130 detected at nearly identical rates regardless of position.

## 131 5 Discussion

- 132 Our results reveal fundamental trade-offs in deploying small fact-checkers for RAG systems.
- 133 The “lost in the middle” phenomenon is real but dataset-dependent. The U-shaped performance  
134 curve on ExpertQA confirms that small fact-checkers suffer middle-context degradation, with the  
135 subsequent recovery at very long lengths suggesting that the chunking mechanism eventually surfaces  
136 the correct evidence chunk when documents are sufficiently long. On RAGTruth, finer-grained  
137 binning at 250-token intervals reveals a more subtle U-shape for flan-t5-large, with BAcc dropping  
138 from 79.1% (250–500 tokens) to 75.9% (500–750 tokens) before recovering to 78.9% (750–1000  
139 tokens). Bespoke-MiniCheck-7B exhibits a different pattern, peaking in middle bins and degrading at  
140 the longest lengths, which may reflect differences in how the 32k-token context window interacts  
141 with mid-length documents. The varying U-shape strength across datasets underscores that context  
142 degradation is influenced by document length distribution, evidence placement, and model architecture  
143 jointly.
- 144 The accuracy-throughput trade-off is severe: the 250× throughput reduction on SummHay highlights  
145 the operational cost of achieving perfect accuracy on extremely long documents via exhaustive  
146 chunking. In production RAG systems, this may necessitate adaptive chunking strategies that scale  
147 processing with document length or confidence thresholds.
- 148 Adversarial hallucinations systematically bypass detection. The 57.5% overall fooling rate, with  
149 entity hallucinations achieving 0.0% conditional detection on RAGTruth, demonstrates that fabricated  
150 citations and fake entities completely evade the chunking-based verification mechanism. This failure  
151 mode is particularly concerning for scientific and legal applications where entity verification is critical.  
152 Approaches like chain-of-verification [Dhuliawala et al., 2023] and self-checking [Li et al., 2023]  
153 may offer paths to mitigate such failures, though their effectiveness under long-context conditions  
154 remains to be evaluated.
- 155 Critically, we find no positional defense against adversarial injection: detection rates are nearly  
156 identical across beginning, middle, and end positions. This rules out simple heuristics such as biasing

157 attention toward document boundaries and suggests that more sophisticated verification strategies,  
158 potentially involving cross-chunk consistency checks, are needed.

159 Several limitations warrant consideration. Our adversarial injection uses synthetic hallucinations  
160 which, while controlled and reproducible, may not capture the full diversity of real-world hallucination  
161 strategies. OpenRouter model evaluations used small sample sizes ( $n=10-50$ ) due to API cost  
162 constraints, limiting the statistical power of those comparisons. The U-shape analysis is constrained  
163 by dataset composition: only ExpertQA provides samples across all five length bins, and TofuEval  
164 documents occupy a narrow token range (500–1,000) that precludes observing a full U-shaped curve.  
165 Future work should extend evaluation to datasets with more uniform length coverage and explore  
166 mitigation strategies such as hierarchical chunking or cross-chunk attention mechanisms.

## 167 6 Conclusion

168 We evaluated MiniCheck-style small fact-checkers under long-context and adversarial stress, revealing  
169 several key findings. First, the “lost in the middle” phenomenon is confirmed on ExpertQA, with  
170 a 9–10% BAcc degradation at 1k–2k tokens followed by recovery at longer lengths; finer binning  
171 on RAGTruth reveals a more subtle U-shape pattern, though its strength varies across models.  
172 Second, extreme-context accuracy incurs a severe throughput penalty, with a 250× slowdown on  
173 74k-token documents, underscoring the need for adaptive chunking in production deployments. Third,  
174 adversarial hallucinations fool the models 57.5% of the time overall, with fabricated entities bypassing  
175 detection entirely on RAGTruth, a finding with serious implications for fact-checking in high-stakes  
176 domains. Finally, injection detection is position-invariant, ruling out simple boundary-biasing  
177 defenses and motivating the development of cross-chunk verification strategies. Taken together,  
178 these results establish that while small fact-checkers offer a practical trade-off between accuracy and  
179 efficiency for RAG pipelines, their vulnerability to adversarial content and context-length degradation  
180 demands careful architectural consideration in real-world deployments.

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